Practice workbook · Describe DNA's double helix, base pairing and replication

Two strands linked by complementary bases and hydrogen bonds, deducing bases or proportions, and copying DNA with DNA polymerase before division.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DNA-STRUCTURE-AND-REPLICATION

1. Question 3 · part c · 4 marks

June 2024 Paper 1

Worked-example retry · Attempt part c only.

3 (a) Polydactyly is an inherited condition.

There are several faulty genes on different chromosomes that can cause polydactyly.

(i) State which part of a cell contains the most chromosomes.

(1)

(ii) Genes exist in alternative forms.

State the name given to an alternative form of a gene.

(1)

(iii) Give one symptom of polydactyly.

(1)

(b) Polydactyly is caused by the dominant form of the gene.

The table shows the genotypes of two parents, R and S.

Parent	Genotype
female (R)	Pp
male (S)	pp

(i) Name the term used to describe the genotype of parent R.

(1)

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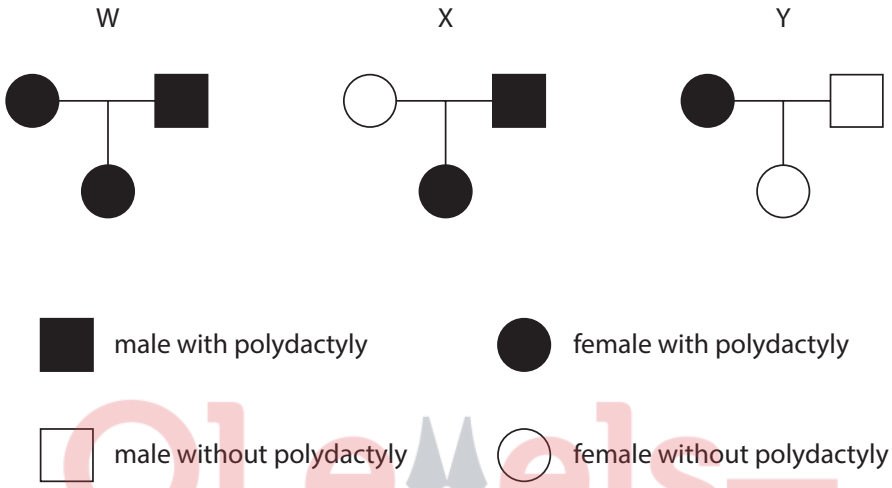
1. Question 3 · part c · 4 marks

June 2024 Paper 1

Worked-example retry · Attempt part c only.

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(ii) The family pedigree diagrams show the inheritance of polydactyly.



Which of the diagrams, W, X or Y, gives the family pedigree for parents R and S?

(1)

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(c) A gene is a short section of DNA.

Describe the structure of DNA.

(4)

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(Total for Question 3 = 9 marks)

1. Question 3 · part c · 4 marks · Working and answer

Working and answer

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2. Question 1 · part a.iii · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

1 (a) The diagram shows part of a molecule of DNA.



(i) Structure X is the backbone of the DNA molecule.

Which of these form part of the backbone of DNA?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

(ii) What name is given to the structures labelled Y?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

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P 7 0 9 7 3 A 0 2 2 4

2. Question 1 · part a.iii · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

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(iii) Describe how the two backbones of the DNA molecule are joined together.

(2)

(iv) State the name of the part of the cell where most of the DNA is found.

(1)

(b) DNA is a nucleic acid.

RNA is another type of nucleic acid.

(i) Give two differences between the structures of DNA and RNA.

(2)

1

2

(ii) There are different types of RNA.

Which of these are two types of RNA?

(1)

- ☒ **A** mRNA and pRNA
- ☒ **B** dRNA and tRNA
- ☒ **C** dRNA and pRNA
- ☒ **D** mRNA and tRNA

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2. Question 1 · part a.iii · 2 marks · Working and answer

Working and answer

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3. Question 1 · part b.ii · 3 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

1 (a) The table shows some cell structures and their functions.

(i) Place one tick (✓) in each row to show the function of each cell structure.

(3)

Cell structure	Function		
	makes proteins	**contains DNA**	**allows substances in and out of cells**
nucleus			
ribosomes			
cell membrane			

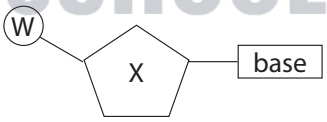
(ii) Give a structure found in bacterial cells.

(1)

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(b) (i) DNA is made of two chains of nucleotides.

The diagram shows one nucleotide.



Name part W and part X.

(2)

W .....

X .....

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P 7 2 5 8 8 A 0 2 2 4

3. Question 1 · part b.ii · 3 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

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(ii) Describe how bases hold the two chains of nucleotides together.

(3)

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(c) The gene for the hormone insulin contains 153 bases.

Determine the number of amino acids in insulin.

(2)

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number of amino acids = ...

(Total for Question 1 = 11 marks)

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3. Question 1 · part b.ii · 3 marks · Working and answer

Working and answer

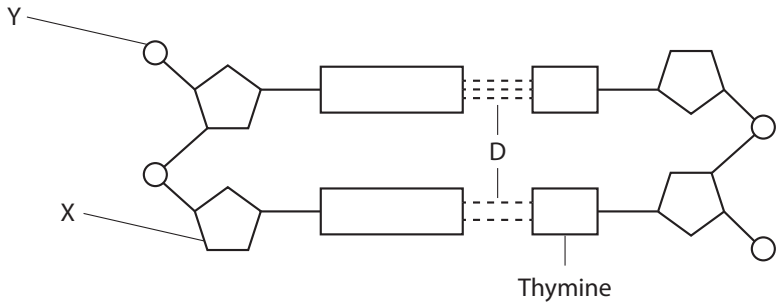
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4. Question 2 · part b.i · 1 mark

June 2025 Paper 1

Further practice · Attempt part b.i only.

2 The diagram shows part of a molecule of nucleic acid.



(a) (i) Name the parts labelled X and Y.

(2)

X .....

Y .....

(ii) Name the type of nucleic acid shown in the diagram.

(1)

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(iii) Use information from the diagram to justify your answer to part (ii).

(3)

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(iv) State where this nucleic acid is found in a human cell.

(1)

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P 7 8 7 6 8 A 0 4 2 0

4. Question 2 · part b.i · 1 mark

June 2025 Paper 1

Further practice · Attempt part b.i only.

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(b) (i) State what is labelled D in the diagram.

(1)

(ii) Describe the importance of D in maintaining the structure of the nucleic acid.

(3)

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(Total for Question 2 = 11 marks)

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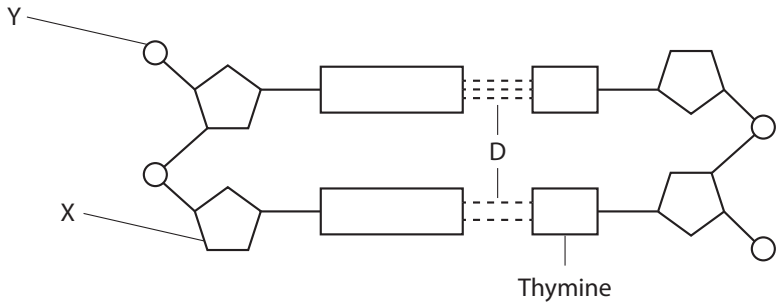
Working and answer

5. Question 2 · part b.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part b.ii only.

2 The diagram shows part of a molecule of nucleic acid.



(a) (i) Name the parts labelled X and Y.

(2)

X .....

Y .....

(ii) Name the type of nucleic acid shown in the diagram.

(1)

.....

(iii) Use information from the diagram to justify your answer to part (ii).

(3)

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(iv) State where this nucleic acid is found in a human cell.

(1)

.....

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P 7 8 7 6 8 A 0 4 2 0

5. Question 2 · part b.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part b.ii only.

(b) (i) State what is labelled D in the diagram.

(1)

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(ii) Describe the importance of D in maintaining the structure of the nucleic acid.

(3)

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(Total for Question 2 = 11 marks)

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5. Question 2 · part b.ii · 3 marks · Working and answer

Working and answer

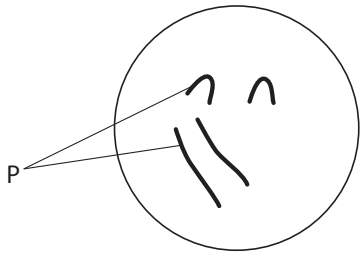
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6. Question 8 · part c.i · 1 mark

November 2020 Paper 1

Further practice · Attempt part c.i only.

8 (a) The diagram shows the nucleus of a cell.



(i) Name the structures labelled P.

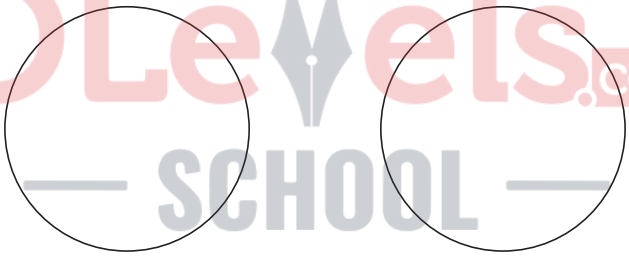
(1)

.....

(ii) The cell divides by mitosis and two nuclei are produced.

Draw the appearance of the two nuclei.

(2)



(b) Before mitosis can occur, the DNA of the cell must replicate.

The table lists stages of DNA replication.

Write the numbers 1, 2, 3 and 4 in the boxes to show these stages in the correct order.

(1)

Stage	Number
complementary bases pair up	
DNA double helix unwinds	
strands separate	
two DNA helices form	

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6. Question 8 · part c.i · 1 mark

November 2020 Paper 1

Further practice · Attempt part c.i only.

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(c) The DNA of the nucleus is analysed.

The table shows the results.

Base	X	adenine	Y	Z
Pecentage base in DNA (%)	30	20	30	20

(i) Identify base Z.

(1)

(ii) Give the possible identities of base X.

(1)

(d) Meiosis is another type of cell division.

Explain why meiosis is important in humans.

(4)

(Total for Question 8 = 10 marks)

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6. Question 8 · part c.i · 1 mark · Working and answer

Working and answer

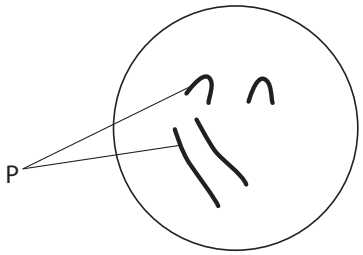
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7. Question 8 · part c.ii · 1 mark

November 2020 Paper 1

Further practice · Attempt part c.ii only.

8 (a) The diagram shows the nucleus of a cell.



(i) Name the structures labelled P.

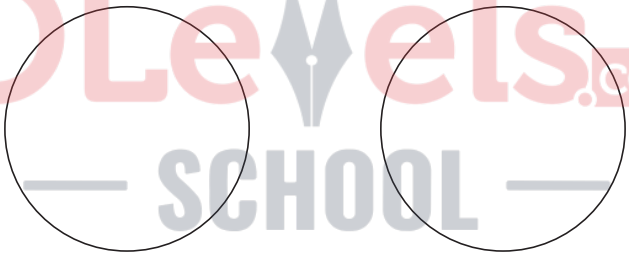
(1)

.....

(ii) The cell divides by mitosis and two nuclei are produced.

Draw the appearance of the two nuclei.

(2)



(b) Before mitosis can occur, the DNA of the cell must replicate.

The table lists stages of DNA replication.

Write the numbers 1, 2, 3 and 4 in the boxes to show these stages in the correct order.

(1)

Stage	Number
complementary bases pair up	
DNA double helix unwinds	
strands separate	
two DNA helices form	

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7. Question 8 · part c.ii · 1 mark

November 2020 Paper 1

Further practice · Attempt part c.ii only.

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(c) The DNA of the nucleus is analysed.

The table shows the results.

Base	X	adenine	Y	Z
Pecentage base in DNA (%)	30	20	30	20

(i) Identify base Z.

(1)

(ii) Give the possible identities of base X.

(1)

(d) Meiosis is another type of cell division.

Explain why meiosis is important in humans.

(4)

(Total for Question 8 = 10 marks)

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7. Question 8 · part c.ii · 1 mark · Working and answer

Working and answer

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8. Question 6 · part b · 2 marks

November 2020 Paper 1R

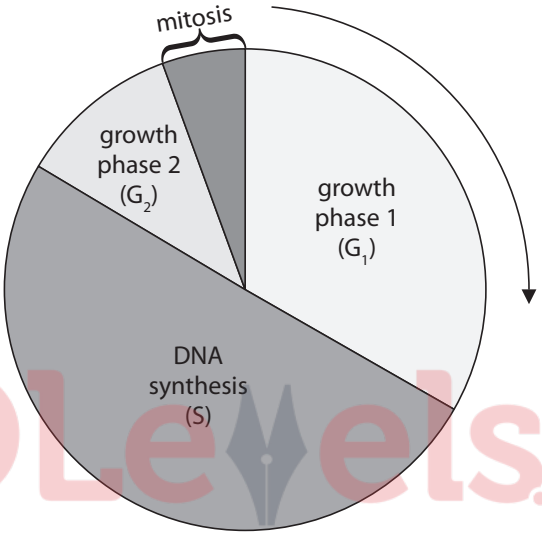
Further practice · Attempt part b only.

6 (a) Cells in the body go through several cell cycles in their lifetime.

Each cell cycle is divided into different stages.

Different events take place in the cell during each stage.

The diagram shows one cell cycle of a body cell.



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(i) One cell cycle takes 24 hours to complete.

The total time spent in G_1 and G_2 is ten hours.

Calculate the time spent in mitosis.

(2)

time = ... hours

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8. Question 6 · part b · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

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(ii) Each of the phases of the cell cycle prepare the cell for mitosis.

Suggest the events that take place in the growth phases of the cell cycle.

(2)

(iii) The cell cycle completes by producing two new body cells.

Describe the genetic features of the two cells produced by mitosis.

(2)

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(iv) The table lists some of the stages of mitosis.

Complete the table by giving details of the event that takes place in each stage.

(3)

Stage	Events
metaphase	
anaphase	
telophase	

8. Question 6 · part b · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

(b) Describe the role of DNA polymerase in the synthesis of new DNA molecules.

(2)

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(Total for Question 6 = 11 marks)

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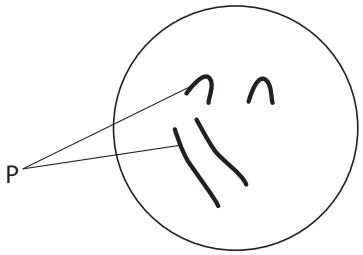
Working and answer

9. Question 8 · part b · 1 mark

November 2020 Paper 1

Further practice · Attempt part b only.

8 (a) The diagram shows the nucleus of a cell.



(i) Name the structures labelled P.

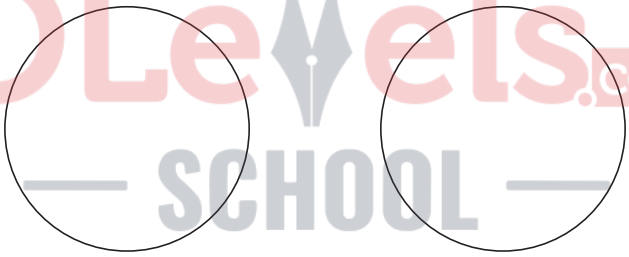
(1)

.....

(ii) The cell divides by mitosis and two nuclei are produced.

Draw the appearance of the two nuclei.

(2)



(b) Before mitosis can occur, the DNA of the cell must replicate.

The table lists stages of DNA replication.

Write the numbers 1, 2, 3 and 4 in the boxes to show these stages in the correct order.

(1)

Stage	Number
complementary bases pair up	
DNA double helix unwinds	
strands separate	
two DNA helices form	

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9. Question 8 · part b · 1 mark

November 2020 Paper 1

Further practice · Attempt part b only.

(c) The DNA of the nucleus is analysed.

The table shows the results.

Base	X	adenine	Y	Z
Pecentage base in DNA (%)	30	20	30	20

(i) Identify base Z.

(1)

(ii) Give the possible identities of base X.

(1)

(d) Meiosis is another type of cell division.

Explain why meiosis is important in humans.

(4)

(Total for Question 8 = 10 marks)

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9. Question 8 · part b · 1 mark · Working and answer

Working and answer

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10. Question 4 · part b.i · 4 marks

November 2023 Paper 1

Further practice · Attempt part b.i only.

4 The mass of DNA in a body cell changes during the cell cycle.

(a) (i) Which structure in a cell contains DNA?

(1)

- ☒ **A** chromosome
- ☒ **B** endoplasmic reticulum
- ☒ **C** cell membrane
- ☒ **D** ribosome

(ii) Name **one** other type of nucleic acid.

(1)

.....

(iii) Give **three** differences between DNA and the nucleic acid that you have named in (a)(ii).

(3)

Differences	DNA	Named nucleic acid
1		
2		
3		

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10. Question 4 · part b.i · 4 marks

November 2023 Paper 1

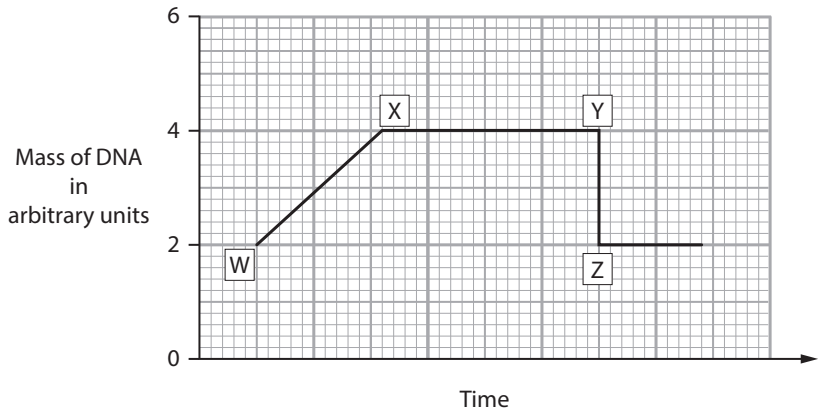
Further practice · Attempt part b.i only.

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(b) The graph shows the changes in the mass of DNA in a body cell, during mitosis.



(i) Describe the process that produces the change in the mass of DNA between points W and X.

(4)

(ii) Describe what is happening in the cell between points Y and Z.

(2)

10. Question 4 · part b.i · 4 marks · Working and answer

Working and answer

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